Figure 1.

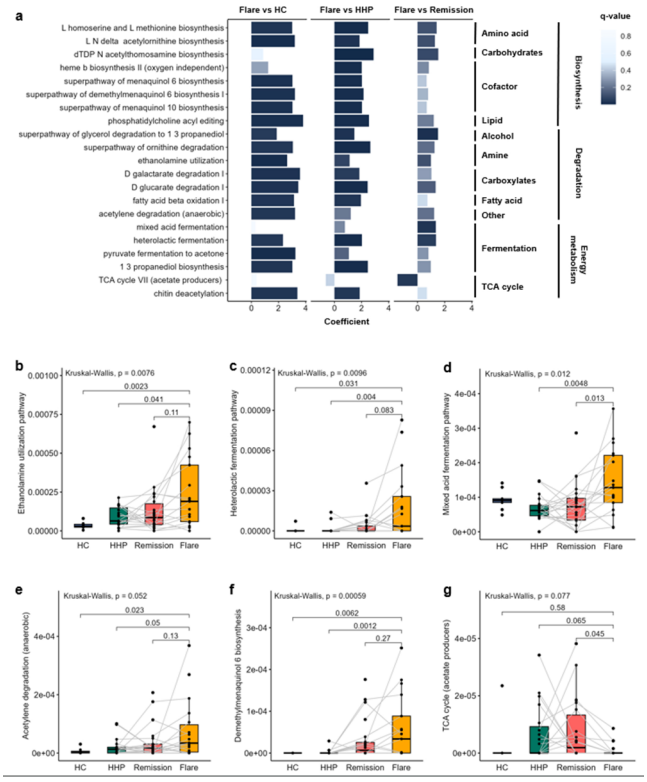


Figure 1. (A) Barplot of significant associations detected by MaAsLin 2 and corresponding coefficients representing differential association strength of specific metabolic pathways between flare and healthy controls (HC), household partners (HHP), or remission, respectively. Only metabolic pathways with significant associations (q-value < 0.05 between Flare and Remission or q-value < 0.02 between Flare and HHP or Flare and HC) are shown. All detected associations are adjusted for households as random effects and disease status as fixed effects. (B-G) Abundance of different metabolic pathways: ethanolamine utilization pathway (B), heterolactic fermentation pathway (C), mixed acid fermentation pathway (D), acetylene degradation pathway (E), demethylmenaquinol 6 biosynthesis pathway (F), and TCA cycle (acetate producers) (G). Grey lines connect samples within the same household.

Figure 2.

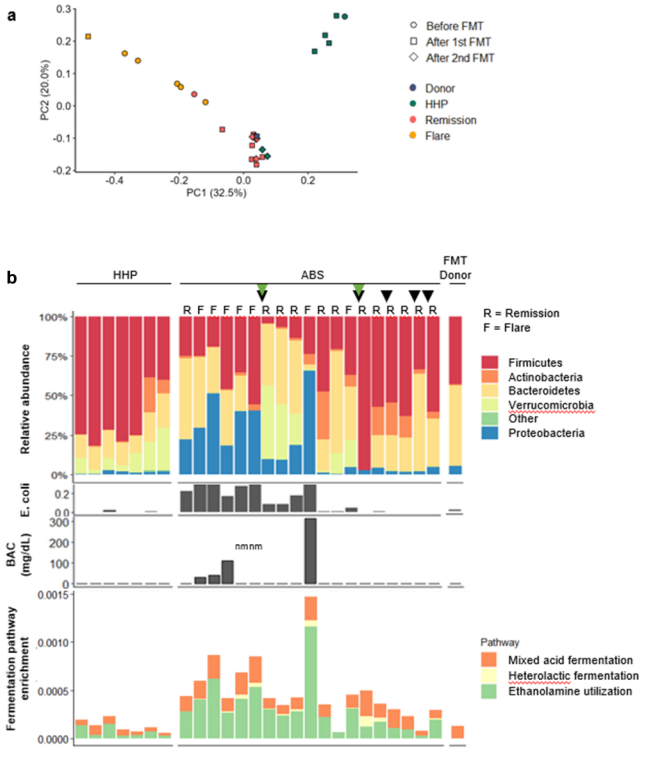


Figure 2. Gut microbial composition after fecal microbiota transplantation. (A) Principal component analysis of gut microbial taxonomic composition at the species level of one patient and their household partner (HHP) before, during, and after two different treatment courses of fecal microbiota transplantation (FMT), during remission or flare. (B) Relative abundance of gut microbiota at the phyla level, relative abundance of *Escherichia coli*, blood alcohol concentration, and fermentation pathway enrichment of samples in (A) across time. nm for BAC = not measured. Green arrows indicate broad spectrum antibiotics and black arrows indicate FMT capsule administration.

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METATRANSCRIPTOMICS REVEALS BACTERIAL TRANSGENES CONFERRING THE BENEFICIAL METABOLIC EFFECTS OF TIME-RESTRICTED FEEDING

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BACKGROUND: Diurnal rhythmicity in the gut plays a crucial role in maintaining gut integrity, circadian rhythms, and metabolic homeostasis. However, existing studies have primarily focused on microbial composition rather than their transcriptional activity, limiting our understanding of microbial functions. Metatranscriptomics helps us understand how microbes behave within a host, revealing biochemical programs that are nearly impossible to replicate *in vitro*. The metatranscriptome is also highly responsive to environmental changes and thereby offers a more dynamic view of microbial functions than gene content alone. Time-restricted feeding (TRF) is a behavioral intervention that prevents deleterious effects of diet-induced obesity (DIO). Given metatranscriptomics' sensitivity to temporal changes, we hypothesized metatranscriptomics would capture time dependent microbial shifts induced by TRF and help identify specific microbial functions that could be mediating beneficial effects once engineered into live bacterial therapeutics. METHODS: To explore microbial functional dynamics, we characterized diurnal fluctuations in the mouse cecal metatranscriptome and metagenome of mice given a high-fat diet ad libitum and 8-hour TRF. Mice under normal chow ad libitum were also used as controls. These mice were put under these regimens for 8 weeks and at the end of this period, cecal content was collected every 4 hours over 24h. Once candidate genes were identified that capture the beneficial effects of TRF, we introduced these genes into a persistently engrafting, native *Escherichia coli* chassis to assess the response of these engineered bacteria *in vitro* and *in vivo*. RESULTS: In this study we show that metatranscriptomics uncovers time-dependent microbial functional shifts induced by TRF that remain undetectable with metagenomics alone. This included TRF recovering cycling and diurnal patterns lost under DIO but only when examining the metatranscriptomics transcripts, not metagenomic genes. We also discovered that the bile salt hydrolase (BSH), a microbial enzyme that deconjugates bile acids, from *Dubosiella newyorkensis* exhibits unique diurnal expression in the TRF group. Engineering this BSH, along with other candidates, into a native *E. coli* chassis revealed distinct differences in deconjugation and amidation activities, underscoring the functional specificity among BSHs. Out of these new engineered microbes, the NKD7 *D. newyorkensis* BSH improved insulin sensitivity, glucose tolerance, and body composition of mice *in vivo*. CONCLUSIONS: These findings suggest that *D. newyorkensis* BSH is integral to the metabolic benefits of TRF and highlight metatranscriptomics as a powerful approach for uncovering time-dependent

bacterial functions related to health and disease, particularly when combined with engineered bacterial systems.

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DIETARY CHOLESTEROL-INDUCED MICROBES TRIGGER IMMUNE RESPONSES IN METABOLIC DYSFUNCTION-ASSOCIATED STEATOHEPATITIS

Jake B. Hermanson, Md Amran Gazi, Malena Cid de la Paz, Trinidad Fernandez Wallace, Paul Kuehnert, Arion J. Kennedy, Lautaro Rostoll Cangiano, Claire O'Leary, Federico E. Rey, Vanessa A. Leone

Metabolic dysfunction-associated steatotic liver disease (MASLD) affects over a third of the global population. The more advanced form, metabolic dysfunction-associated steatohepatitis (MASH) will soon become the leading cause of liver transplant. With limited FDA-approved interventions, there is an urgent need for therapies to promote reversal of fibrosis. Data in humans links high dietary cholesterol (HDC) to worse MASLD outcomes. Specific pathogen-free (SPF) mice harboring microbes fed HDC develop fibrosing MASH while germ-free (GF) counterparts lacking microbes are protected. This suggests HDC-induced microbes are required for MASLD onset and progression to fibrosing MASH. Connections between gut microbiota translocation to the liver and the consequences for hepatic immune dysregulation in the context of MASH progression to fibrosis remain unclear. Gaining insight into the mechanistic underpinnings of these interactions may identify immunological targets to combat fibrosing MASH. We hypothesized translocation of specific gut microbes induced by HDC intake would drive early shifts in hepatic antigen-presenting cells (APC) that contribute to aberrant $CD4^+$ and $CD8^+$ T cell responses, which leads to MASH with fibrosis. Male 16-week-old GF and SPF mice were fed high-fat (HF) or HF+2% cholesterol (HFHC) diets and provided fructose+glucose drinking water for up to 26 weeks. HFHC diet increased plasma Alanine transaminase in SPF mice after 4 weeks ($P<0.01$) and led to a 3-fold increase in hepatic fibrosis based on Picrosirius red staining ($P<0.05$) relative to all groups after 26 weeks. Single-cell RNA sequencing of nonparenchymal cells (NPC) showed enrichment in "PD-L1/PD-1 checkpoint" pathway in HFHC-fed SPF mice (FDR $P<0.01$) relative to all groups. $CD8^+$ and $CD4^+$ T cells upregulated exhaustion markers *Pdcd1* and *Tim3* in HFHC-fed SPF mice compared to all groups ($P<0.05$). This corresponded to increased relative abundance of hepatic Enterobacteriaceae (100% sequence alignment to *Klebsiella pneumoniae*) relative to HF-fed counterparts ($P<0.05$) determined via 16S rRNA gene amplicon sequencing. Examination of NPCs throughout disease course via flow cytometry showed that after 2 weeks, HFHC drove increases in dendritic cells ($P<0.01$) and Kupffer cells ($P<0.01$) relative to HF. This early shift in APCs observed in HFHC-fed mice was followed by increased exhausted PD-1^+CD8^+ T cells ($P<0.05$) after 4, 16, and 20 weeks relative to HF-fed mice, preceding the onset of fibrosis. These data suggest HDC rapidly shifts hepatic APCs, followed by activation and exhaustion of T cells, which may contribute to fibrosis. Whether these shifts in immune cell profiles are mechanistically driven by translocation of gut bacteria, specifically *K. pneumoniae,* will require further exploration, which may aid in developing microbe- and immune-based interventions for fibrosing MASH.

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COMMENSALS DEPLETED IN OBESITY ACTIVATE ENTEROENDOCRINE CELLS AND IMPROVE HOST METABOLISM

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Background: *Blautia* species are relatively common commensals in the human gut, but they are depleted in people with obesity, diabetes, and metabolic syndrome. We previously found that *Blautia* species are less abundant in children with Loss of Control eating, a common form of disordered eating that carries a high risk of developing obesity. Thus, *Blautia* is a key commensal organism that may protect against obesity and metabolic syndrome, but the mechanisms by which it benefits the host are not fully understood. Aim: To delineate the mechanisms by which Blautia benefits host metabolism. Results: We first sought to determine the key metabolites associated with *Blautia* species. To that end, we probed publicly available metagenomic and metabolomic databases and discovered that acyl amines were the class of gut metabolites most highly associated with *Blautia* abundance. Within our unique globally sourced and fully sequenced strain library, we identified several *Blautia* species that express N-acyl transferases, and thus the potential to catalyze acyl amine synthesis. Liquid chromatography-mass spectrometry (LC-MS) analyses of *Blautia* cultures confirmed that these strains produce acyl amines - both *in vitro* and in our newly established mouse model of *Blautia* colonization. Using luciferase-based epithelial cell bioreporters, we then found that *Blautia* acyl amines, particularly the N-acyl dopamines, activate epithelial receptors including TRPV1 and GRP119. Using enteroendocrine cell (EEC) lines and EEC-containing human organoids, we found that purified *Blautia* acyl amines induced secretion of GIP, PYY and GLP-1. In mice, single-dose gavage treatments of *Blautia* acyl dopamines resulted in significantly improved glycemic control, which we assessed using oral glucose tolerance tests. Finally, in two separate human cohorts, we found that the absence or very-low abundance of *Blautia* acyl amine synthesis genes was associated with severe obesity. Conclusions: Altogether, these data support the idea that *Blautia* species may activate enteroendocrine cells to benefit host metabolism and identify a potential microbial target for obesity therapeutics.

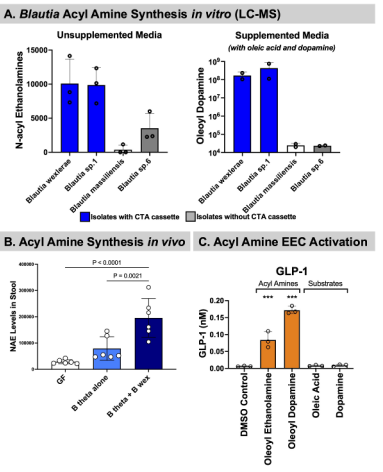


Figure 1.

A. Blautia species encoding the CTA operon (blue) synthesize more acyl amines than *Blautia* isolates without the operon (grey).
Methods: Blautia species were grown in either unsupplemented reinforced clostridial media (RCM) or RCM supplemtned with oleic acid and dopamine (10 μM each). N-acyl ethanolamines and oleoyl dopamine were quantified by LC-MS. Statistical significance calculated by Kruskal Wallis.

B. *B. wexlerae* colonization in mice results in increased increased N-acyl ethanolamines.
Methods: Mice were gavaged with 10^9 C.F.U. of either B. wexlerae or B. theta, or both. Fecal pellets were analyzed by LC-MS. Data are averages of 6 replicates, individual data points shown in addition to SEM.

C. Acyl amines (orange) stimulate GLP-1 secretion from EEC cells. Vehicle controls, oleic acid (lipid precursor) and dopamine (amine precursor) do not stimulate hormone secretion. In data not shown, PYY and GIP are also stimulated in organoid models.
Methods: Purified acyl amines or their precursor substrates (all 10 μM) were used to stimulate NCI-h716 cells. GLP-1 were measured using a luciferase-based reporter. Statistics: T-tests with Dunnett's correction, ***: p < 0.001.

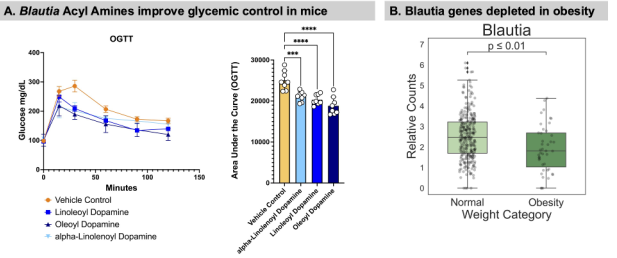


Figure 2.

A. Blautia acyl amines improve glucose tolerance. Left. Oral glucose tolerance test (OGTT) after treatment with *Blautia* acyl amines. Right. Area under the curve of the OGTT.
Methods: Mice were gavaged with 3 mg of acyl amine or vehicle control and an OGTT was performed 30 minutes after. Statistics: T-tests with Dunnett's correction.

B. *Blautia* acyl amine synthesis gene depletion (decreased acyl transferase genes) is associated with obesity.
Methods: Blautia acyl transferase genes were mapped and counted in shotgun metagenomics samples from stools in participants from the Global Microbiome Conservancy (Groussin 2023). Individuals with obesity were compared to normal weighted individuals. Statistics: Mann Whitney U.

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BARIATRIC SURGERY PROMOTES WEIGHT LOSS IN OBESE PATIENTS BY MODULATING GUT MICROBIOTA AND ASSOCIATED METABOLITES

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Objective: Bariatric surgery is a widely recognized intervention for obesity; however, the mechanisms underlying its weight loss effects, beyond reducing food intake, remain poorly understood. This study aimed to investigate whether bariatric surgery promotes weight loss by altering the composition of gut microbiota and associated metabolites. **Design:** A multi-cohort analysis was conducted using metagenomic datasets from four centers, with data collected pre-surgery and at one, three, six, and twelve months post-surgery. Experimental validation was performed using a diet-induced obesity (DIO) mouse model in vivo and an adipogenesis induction model in vitro. **Results:** Comprehensive analysis of multi-center metagenomic datasets revealed significant changes in gut microbiota composition following bariatric surgery, with notable enrichment of butyrate-producing bacteria and associated metabolic pathways. The abundance of *Roseburia hominis* (R. hominis) was significantly increased post-surgery. Interestingly, lower levels of R. hominis were observed in obese individuals compared to healthy controls. Oral gavage of R. hominis significantly attenuated high-fat diet (HFD)-induced obesity in mice, reducing subcutaneous fat, visceral fat, and liver weight. Metabolic assessments showed improved insulin sensitivity and reduced HFD-induced metabolic damage, while histological analysis indicated decreased adipocyte size, suggesting impaired adipogenesis in the R. hominis group. Additionally, conditioned medium from R. hominis significantly suppressed adipogenic differentiation in murine mesenchymal stem cells (MSCs) in vitro. Targeted metabolomics confirmed substantial butyrate production in the R. hominis conditioned medium. Furthermore, oral gavage of butyrate replicated the anti-obesity and metabolic benefits of R. hominis in vivo and inhibited adipogenic differentiation in vitro, identifying butyrate as a key effector molecule of R. hominis. RNA sequencing of R. hominis-conditioned medium-treated adipocytes derived from C3H10 MSCs revealed significant downregulation of adipogenic pathways and activation of the Rap1/MAPK/Erk signaling pathway. This was further validated by increased phosphorylation of MEK and ERK proteins in response to treatment with R. hominis and butyrate. Activation of the Rap1/MAPK/Erk pathway, which suppressed adipogenic differentiation in vitro, suggests that it mediates the anti-obesity effects of R. hominis and butyrate. **Conclusion:** Bariatric surgery promotes weight loss in obese patients by enriching *Roseburia hominis* and butyrate in the gut, which inhibits adipogenesis through activation of the Rap1/MAPK/Erk signaling pathway. These findings provide mechanistic insights into the gut microbiota-mediated metabolic benefits of bariatric surgery and highlight R. hominis and butyrate as potential therapeutic targets for obesity.